

The Transformer Revolution

LIN 313 Language and Computers
UT Austin Fall 2025
Instructor: Gabriella Chronis

Transformer—it's not sci-fi. It's just a name, invented by nerds who like sci-fi (like 'Perceptron').

Objectives

- Understand Machine Translation task
- Read and interpret high-level neural network **architecture** diagrams
- overview of NLP history leading up to LLMs and the main innovations
 - attention

Machine Translation (MT)

- Task: Given an input in a source language (e.g. English), predict the translation into a target language (e.g. French).
- Training data: usually a **parallel corpus** — sentences or documents in the source language paired with their translation.
- the shape of the problem influenced the shape of the models that were developed to solve it
 - MT is a **sequence to sequence** task : given one sequence, produce another
 - other examples: question answering, language modeling . . .



1a. Garcia and associates .	7a. the clients and the associates are enemies .
1b. Garcia y asociados .	7b. los clientes y los asociados son enemigos .
2a. Carlos Garcia has three associates .	8a. the company has three groups .
2b. Carlos Garcia tiene tres asociados.	8b. la empresa tiene tres grupos .
3a. his associates are not strong .	9a. its groups are in Europe .
3b. sus asociados no son fuertes .	9b. sus grupos estan en Europa .
4a. Garcia has a company also .	10a. the modern groups sell strong pharmaceuticals.
4b. Garcia tambien tiene una empresa .	10b. los grupos modernos venden medicinas fuertes .
5a. its clients are angry .	11a. the groups do not sell zenzanine .
5b. sus clientes estan enfadados.	11b. los grupos no venden zanzanina .
6a. the associates are also angry .	12a. the small groups are not modern.
6b. los asociados tambien estan enfadados.	12b. los grupos pequenos no son modernos .

Translate this from English to Spanish:
the modern clients in Europe do not sell pharmaceuticals also .

1990s: Statistical Machine Translation (SMT)

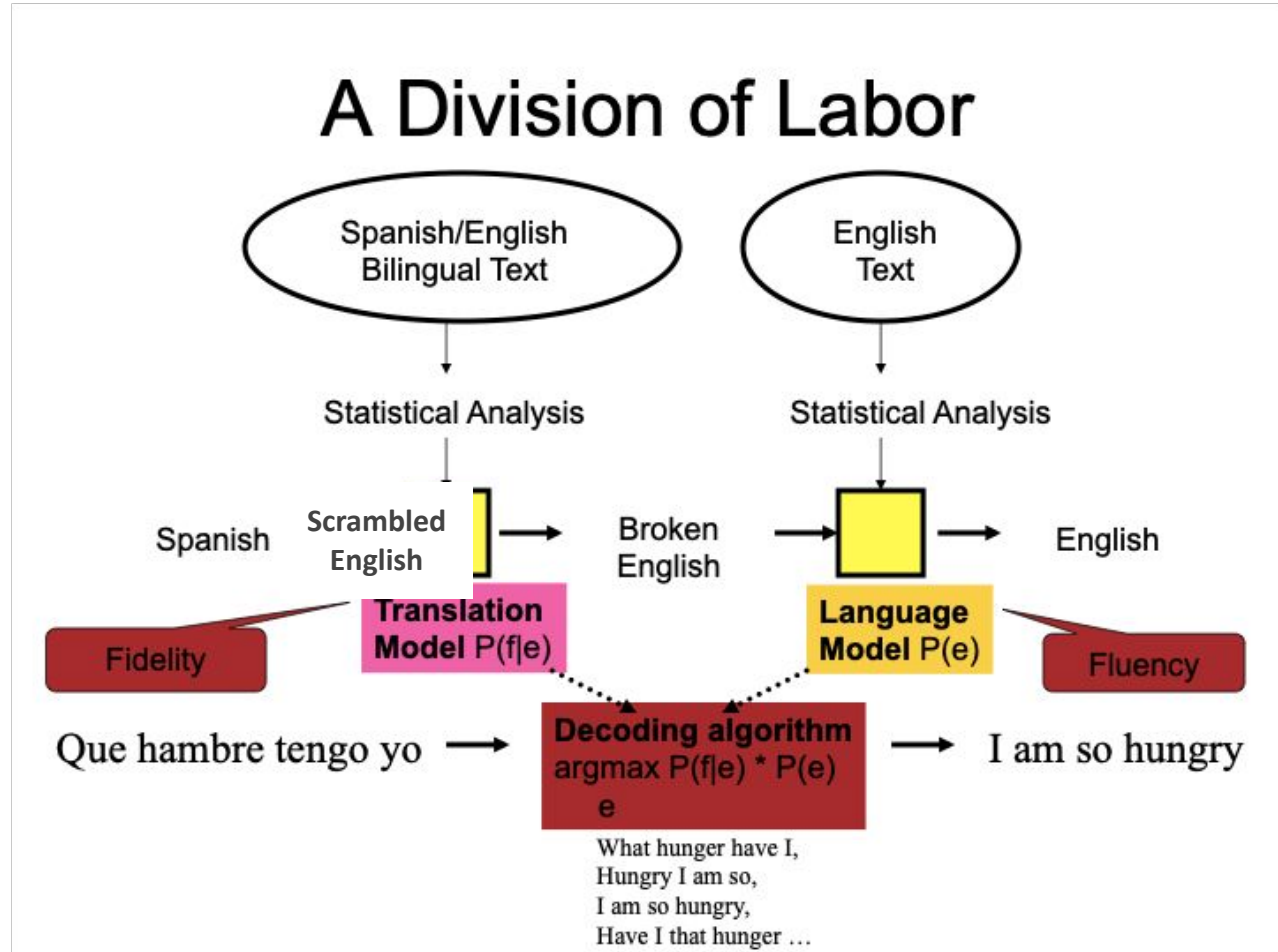
The most popular approach in the 90s **did not use neural networks**.

Statistical machine translation systems are **generative probabilistic models**, like Naive Bayes.

They **calculate** the **probability** of one word in the source text generating a word in the destination language. At translation time, they sample from this distribution.

1990s: SMT

SMT breaks the translation problem up into sub-problems, and then models those subproblems with Bayesian statistics (NOT neural networks)



1990s: Statistical Machine Translation (SMT)

Word Alignment

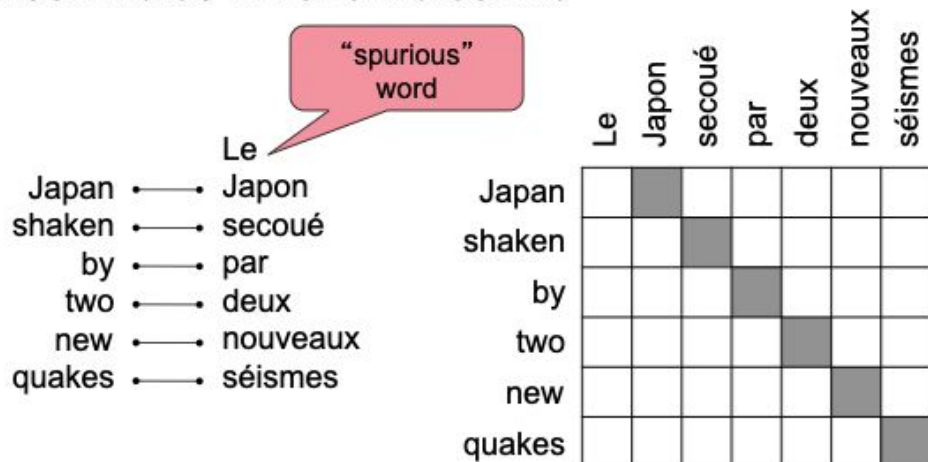
the intuition: break the problem into smaller problems

1. translate words to corresponding words
2. figure out the right order for those words

Part of that problem is "alignment" — figuring out how responsible a word in the input for a word in the output

Alignments

We can factor the translation model $P(f | e)$ by identifying *alignments* (correspondences) between words in f and words in e



Alignments: harder

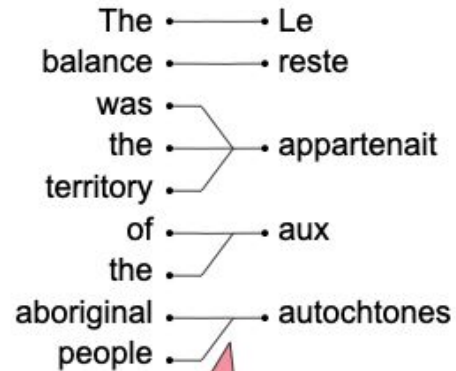
“zero fertility” word
not translated



one-to-many
alignment

	Le	programme	a	été	mis	en	application
And							
the							
program							
has							
been							
implemented							

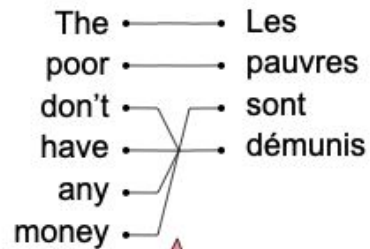
Alignments: harder



many-to-one
alignments

	Le	reste	appartenait	aux	autochtones
The					
balance					
was					
the					
territory					
of					
the					
aboriginal					
people					

Alignments: hardest



many-to-many
alignment

	Les	pauvres	sont	démunis
The				
poor				
don't				
have				
any				
money				

phrase
alignment

2010s: Neural Machine Translation

Researchers tried more and more complex kinds of neural network architectures to try to improve performance.

Recursive Neural Networks (RNNs)

Convolutional Neural Networks (CNNs)

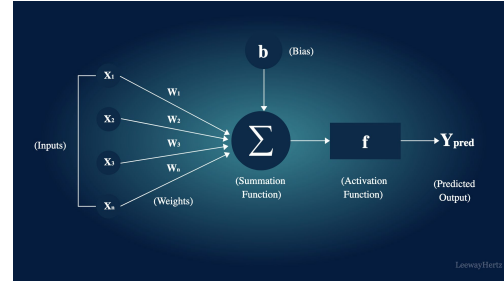
LSTMs (Long-Short-Term Memory networks)

Neural Network Architecture

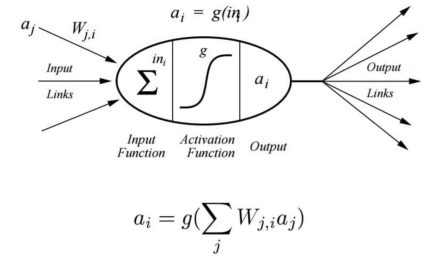
neural network architecture refers to the arrangement of neurons into layers and the connection patterns between layers, activation functions and learning methods.

Architecture diagrams vary widely. They usually show some or all of the following information:

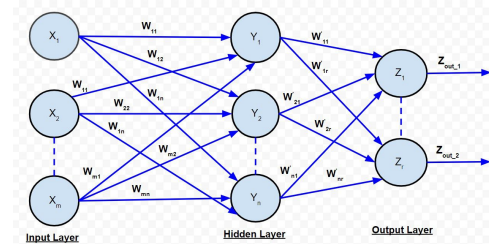
1. what the input looks like
2. what intermediate representations
3. how information flows through the network
4. the different types of functions that information flows through
5. what the output looks like



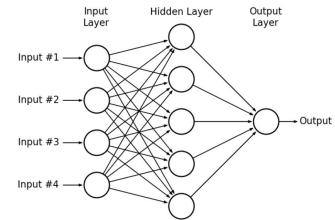
perceptron
architecture



perceptron
architecture



multi-layer perceptron
architecture



multi-layer perceptron
architecture

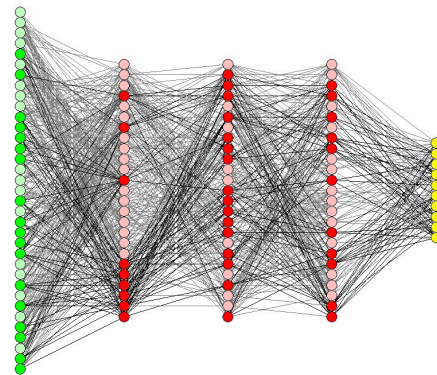
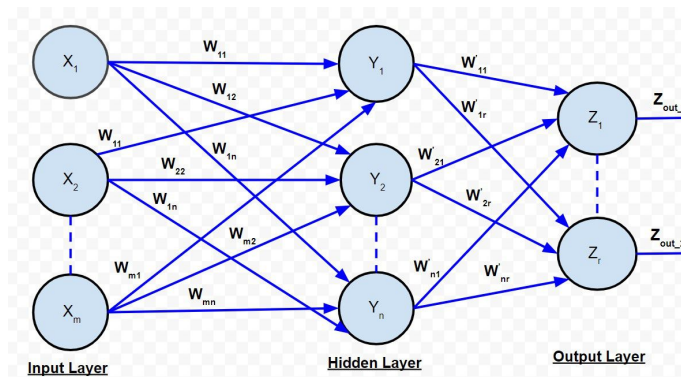
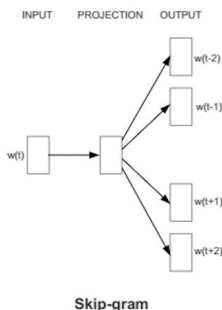
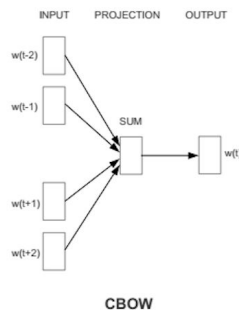
more visualization types:

<https://github.com/ashishpatel26/Tools-to-Design-or-Visualize-Architecture-of-Neural-Network>

Feed Forward Neural Networks (FFNN)

FFNN (Feedforward Neural Network): A neural network where information flows in one direction from input to output through layers, with no cycles or feedback connections.

An FFN is a bunch of perceptrons stitched together



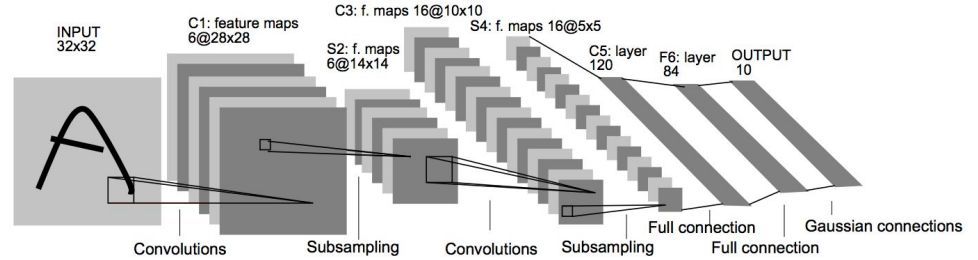
word2vec architectures

Multilayer Perceptron (MLP)

Multilayer
Perceptron (MLP)

2010s: Neural Machine Translation

Convolutional Neural Networks



2010s: Neural Machine Translation

Recurrent Neural Network: (RNN): A neural network with connections that form loops, allowing it to maintain hidden state and process sequential data by feeding outputs back as inputs at each time step.

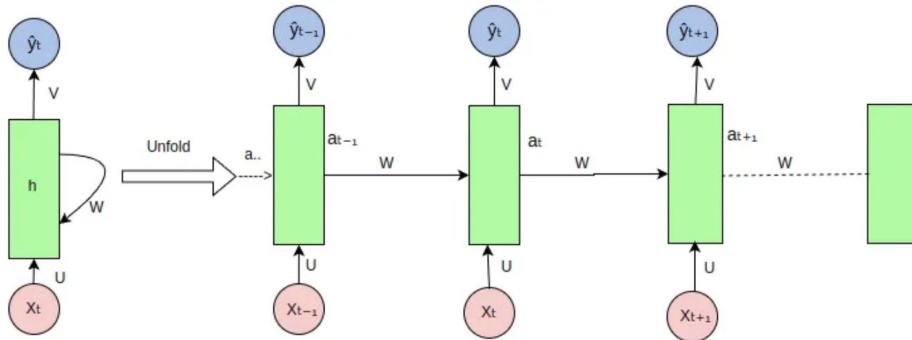
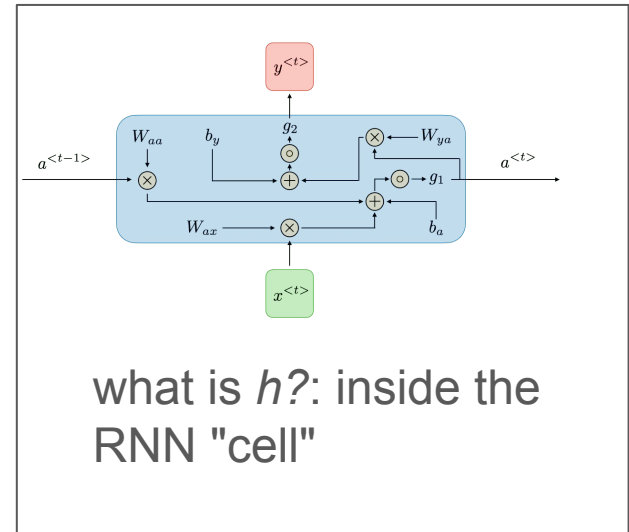
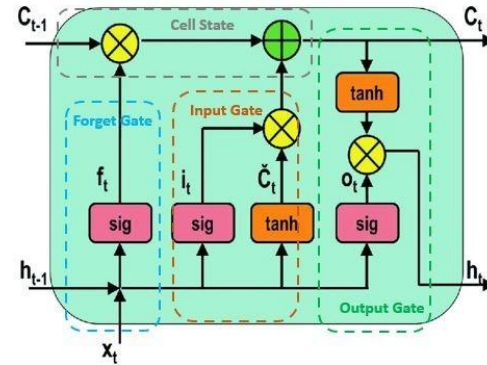


fig 3: RNN Unfolded. Image by Author.



2010s: Neural Machine Translation

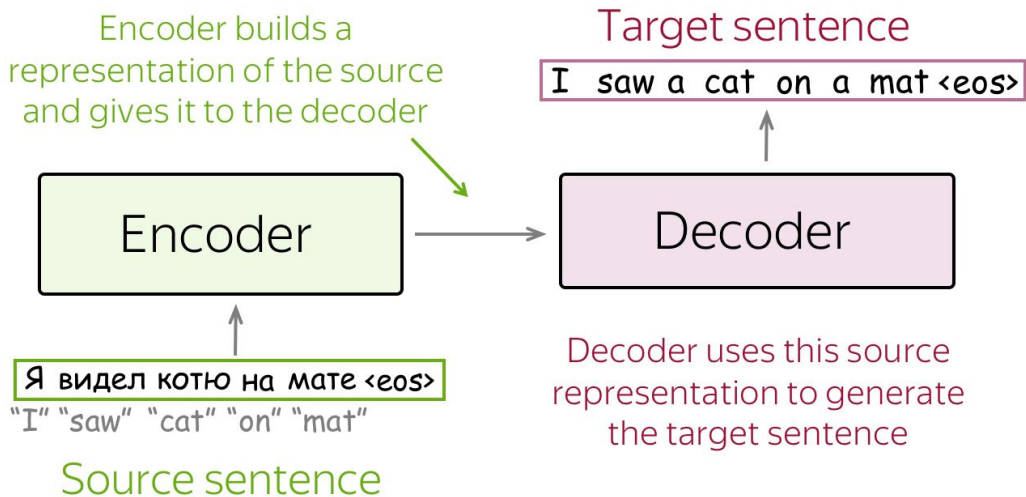
LSTMs (Long-Short Term Memory Networks)



2010s: Neural Machine Translation

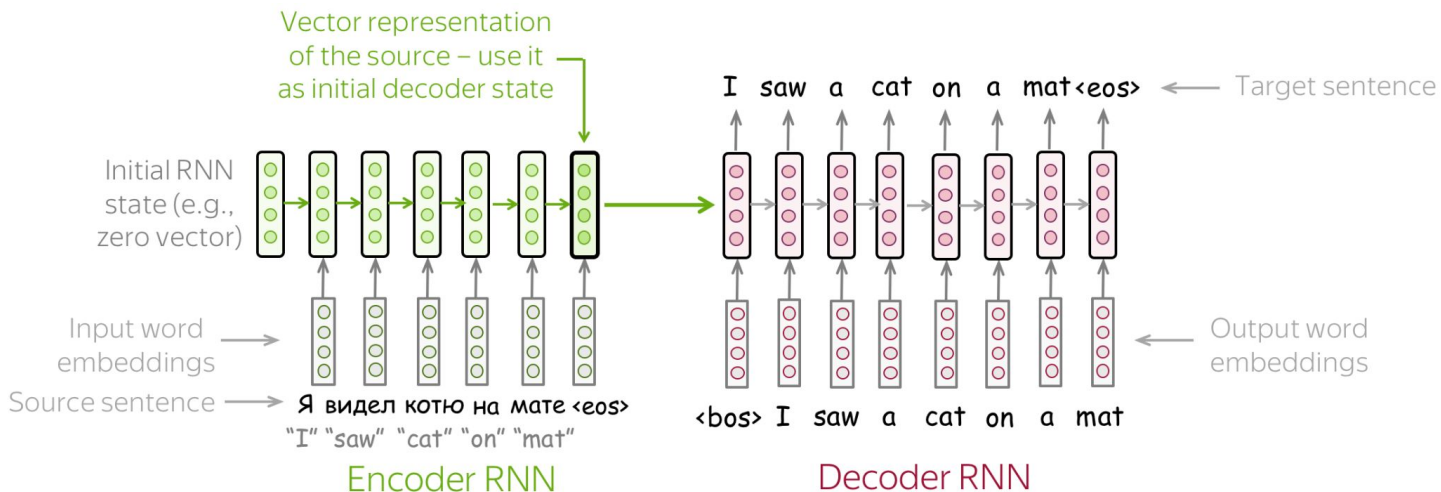
Encoder-Decoder Models

The best translation models were encoder-decoder models

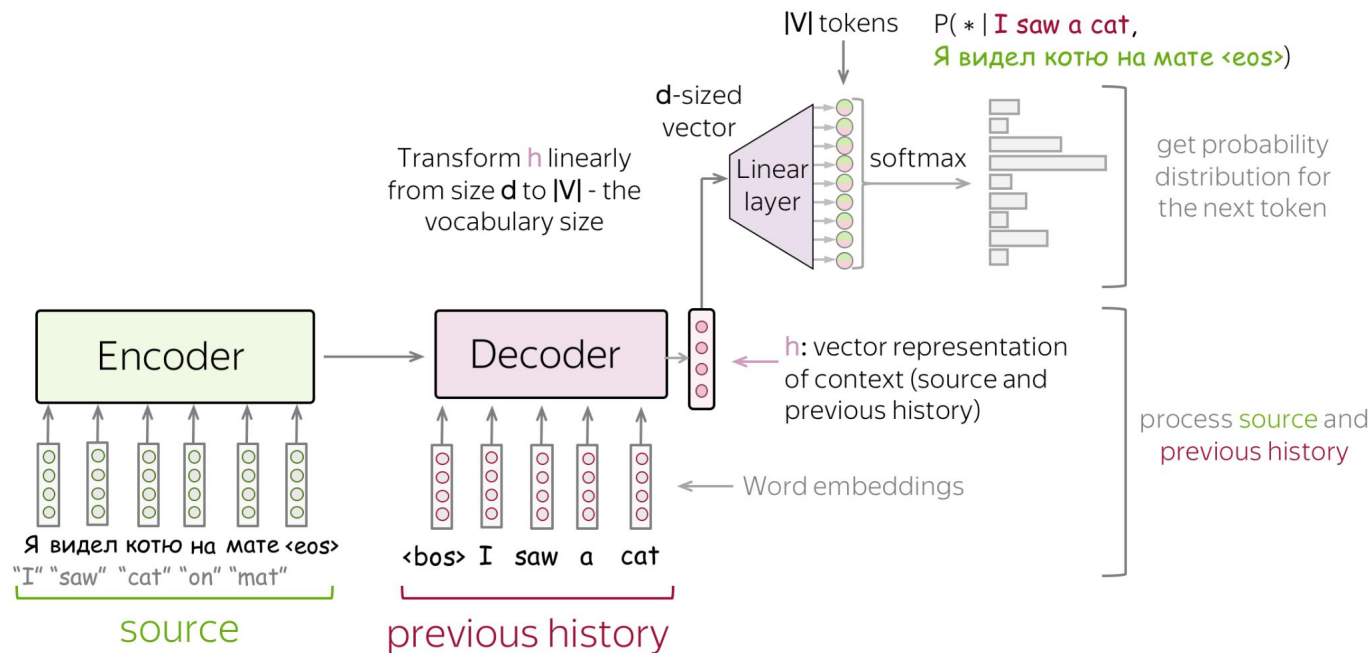


2010s: Neural Machine Translation

A Recurrent Neural Network encoder-decoder model



2010s: Neural Machine Translation



But there is a problem here

To see what it is, think like an encoder decoder.

Read the following sentence (from Roberto Bolaño, 2666):

"The sky, at sunset, looked like a carnivorous flower."

But there is a problem here

To see what it is, think like an encoder decoder.

Read the following sentence (from Roberto Bolaño, 2666):

Now translate the sentence.

But there is a problem here

To see what it is, think like an encoder decoder.

Read the following sentence (from Jose Saramago, *Blindness*):

"On offering to help the blind man, the man who then stole his car, had not, at that precise moment, had any evil intention, quite the contrary, what he did was nothing more than obey those feelings of generosity and altruism which, as everyone knows, are the two best traits of human nature and to be found in much more hardened criminals than this one, a simple car-thief without any hope of advancing in his profession, exploited by the real owners of this enterprise, for it is they who take advantage of the needs of the poor."

But there is a problem here

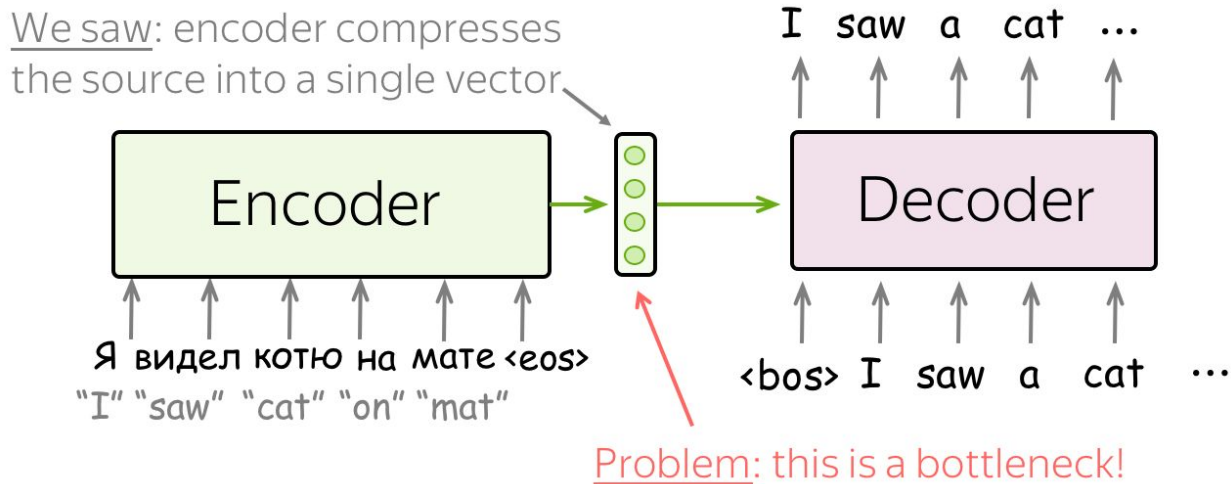
To see what it is, think like an encoder decoder.

Read the following sentence (from Jose Saramago, *Blindness*):

Now translate the sentence.

Encoder-Decoder Models

Problems: (i) for the encoder, it is hard to compress the sentence; (ii) for the decoder, different information may be relevant at different steps.



Attention is All You Need (Vaswani et al. 2017)

- As of 2025, it has been cited more than 173,000 times, placing it among the top ten most-cited papers of the 21st century
- Currently, Transformers (with variations) are de-facto standard models not only in sequence to sequence tasks but also for language modeling and in pretraining settings, which we consider in the next lecture.
- You can do really well at MT with only attention + MLPs (none of that fancy recurrent or LSTM stuff)

Attention Is All You Need

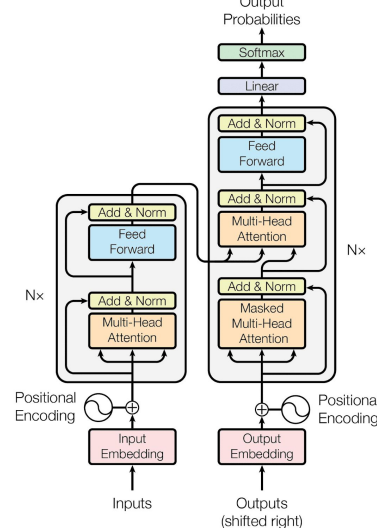
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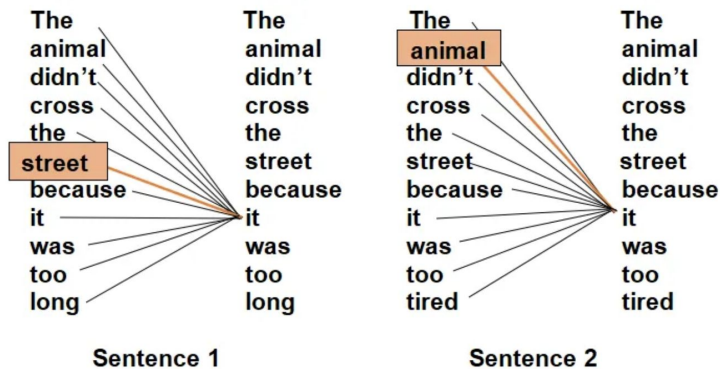


great resource for coding along: <https://nlp.seas.harvard.edu/2018/04/03/attention.html> (the annotated transformer)

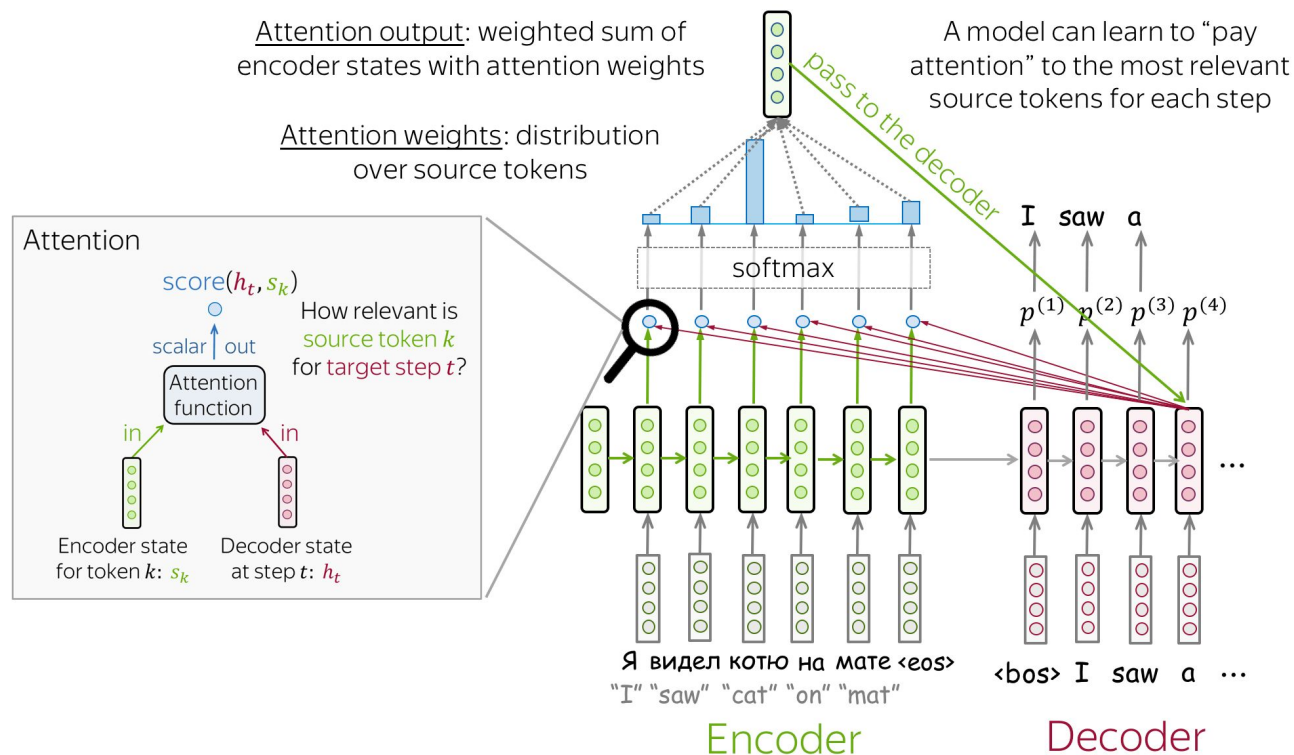
What is Attention?

Intuition: At different steps, let a model "focus" on different parts of the input.

An attention mechanism is a part of a neural network. At each decoder step, it decides which source parts are more important.



What does the attention mechanism "look like"?



To see step-by-step https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

Information flow in the Transformer



What does Attention learn?

From the examples, we see that attention learns (soft) alignment between source and target words - the decoder looks at those source tokens which it is translating at the current step

